

Fraud & Anomaly Detection

Cause-first risk framework using real G'Contest banking data

1,418,030

Transactions scored

109,155

High/Critical cases

27,864

Customers impacted

- No synthetic data and no fake fraud labels.
- Output is a prevention queue with explainable risk reasons.
- Framework aligns with risk-based banking controls: step-up authentication, manual review, AML escalation.
- Optimization priority: catch more fraud-risk cases first, while reporting false-positive rate for customer experience control.

Why Cause-First

BGK note: identify root causes before introducing the model

- Account takeover / identity compromise: new device, new IP, night access, new beneficiary, late-stage digital activity.
- Unauthorized transfer / capital outflow: outside-bank transfer, amount above customer baseline, daily burst, cash-out versus CASA.
- AML network / mule-account pattern: shared IP/device/beneficiary, round high-value transfers, repeated external transfers.

4-Phase Framework

The final system follows the requested 11-step prevention flow

- 1. Foundation** Clean data, create 4 baseline groups, package Customer 360 with rolling 30/60/90 days
- 2. EDA + Labels** Insight charts, IQR thresholding, dynamic root-cause rules, weak labels
- 3. ML + Matrix** RandomForest learns weak labels; Rule+ML matrix converts signals into bank actions
- 4. Operations** Streamlit dashboard, protected amount, SHAP xAI and human-readable reason codes

Prevention Performance

Weak-label evaluation for an operational fraud prevention queue

100.0%

Recall / prevention coverage

100.0%

Precision of Step-up/Block

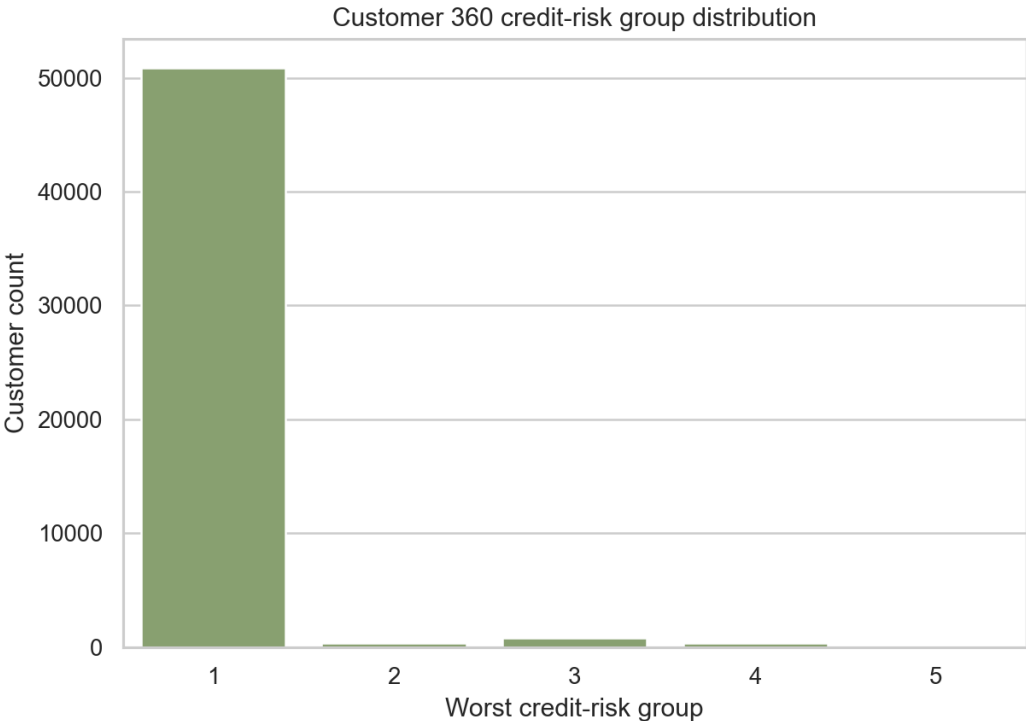
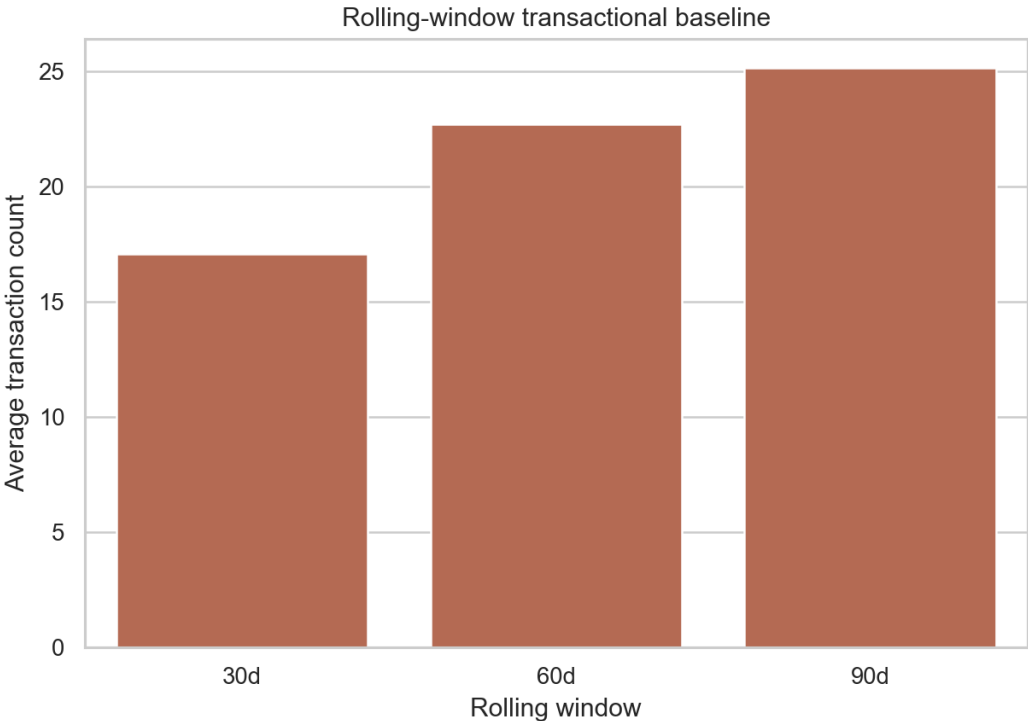
0.00%

False-positive rate

- Metrics are measured against rule-derived weak labels because the contest data has no confirmed fraud labels.
- The bank objective is risk minimization: prioritize catching suspected fraud, then tune thresholds against review capacity and customer friction.
- Protected amount by Block/Step-up queue: 3,418,789,194,096 VND.
- High-threshold confusion matrix: {'true_negative': 43333, 'false_positive': 0, 'false_negative': 2167, 'true_positive': 19500}.

Customer 360 Baseline

One customer, one current financial-behavioral profile



- Transactional: amount average/P95/IQR, golden hour, rolling 30/60/90-day activity.
- Financial: CASA/TD balance, card utilization, overdue and credit-risk group.
- Environmental and behavioral: trusted device/IP, known beneficiary, app activity, night/late-stage activity

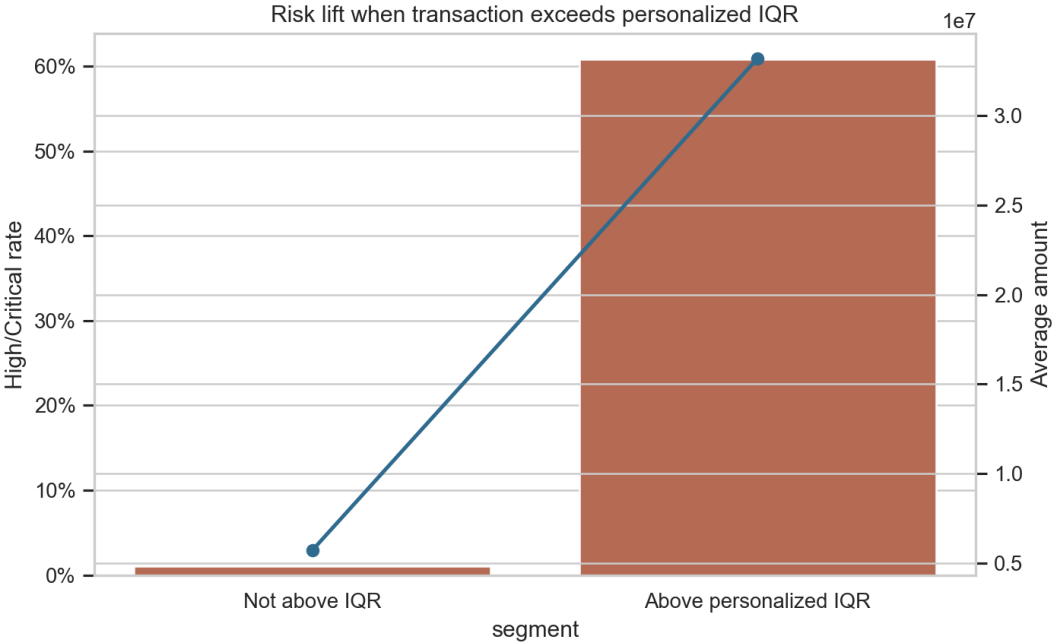
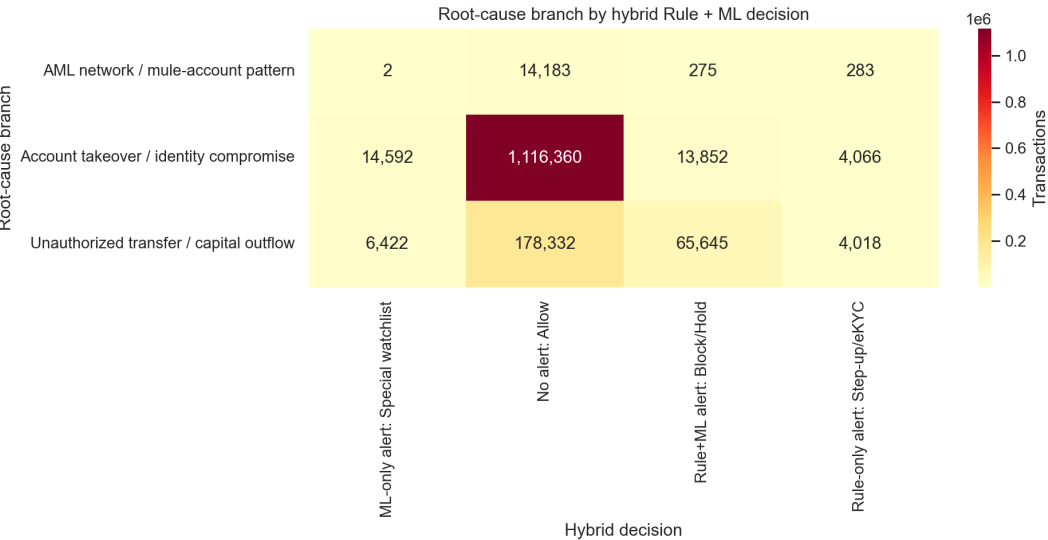
Data-Driven Insights

Charts are selected from actual risk lifts and root-cause interactions

- Root-cause concentration: Unauthorized transfer / capital outflow chiếm 69.7% số giao dịch High/Critical.
- Personalized IQR threshold matters: Giao dịch vượt ngưỡng IQR cá nhân hóa có High/Critical rate 60.8%, lift 57.7x so với nhóm không vượt.
- Network exposure needs a second signal: Nhóm shared IP/device/beneficiary top 1% có High/Critical rate 7.7% so với baseline 7.7%; lift 1.0x.
- Overdue is context, not a fraud label: Nhóm credit-risk group 3-5 có High/Critical rate 7.4%, nhóm 1-2 là 7.7%.

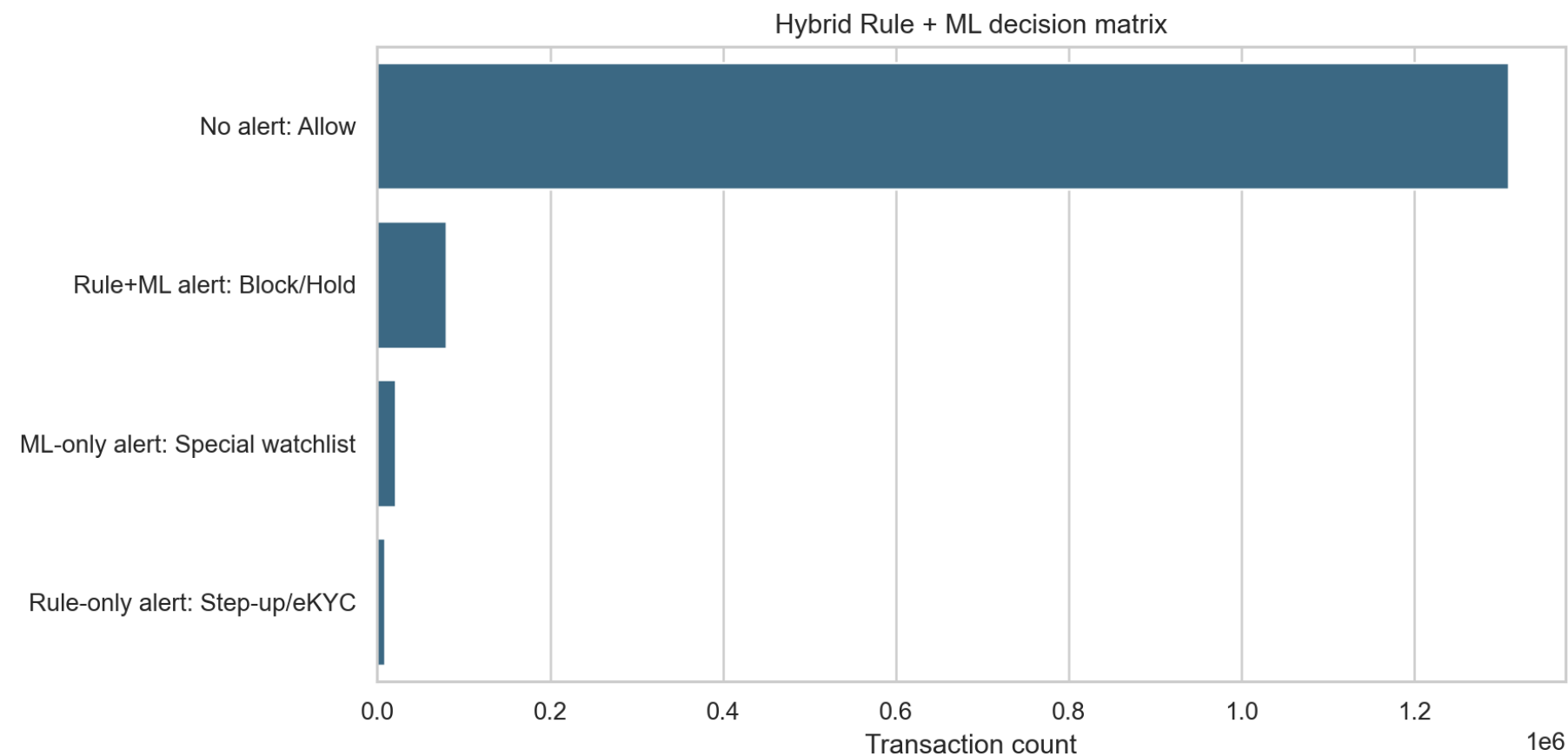
Insight Evidence

Heatmaps and lift charts replace one-dimensional bar-only EDA



Hybrid Decision Matrix

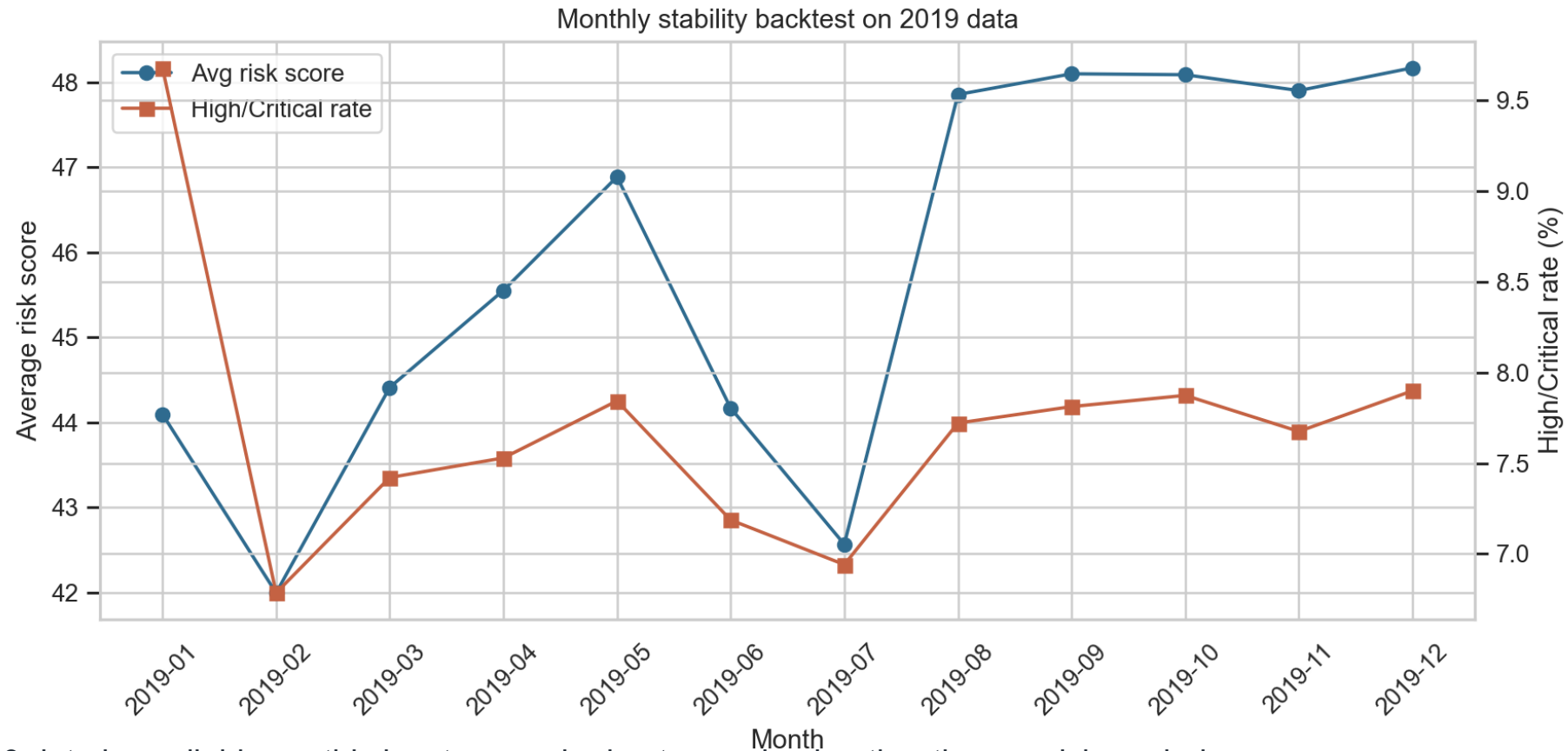
Rule engine and ML confidence are converted into bank actions



- Rule+ML alert: Block/Hold.
- Rule-only alert: Step-up/eKYC to reduce false positives.
- ML-only alert: Special watchlist for patterns not yet covered by rules.
- No alert: Allow while baseline continues to update.

Temporal Stability

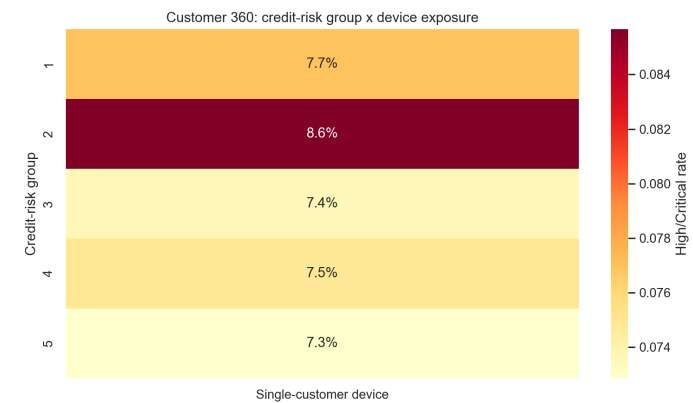
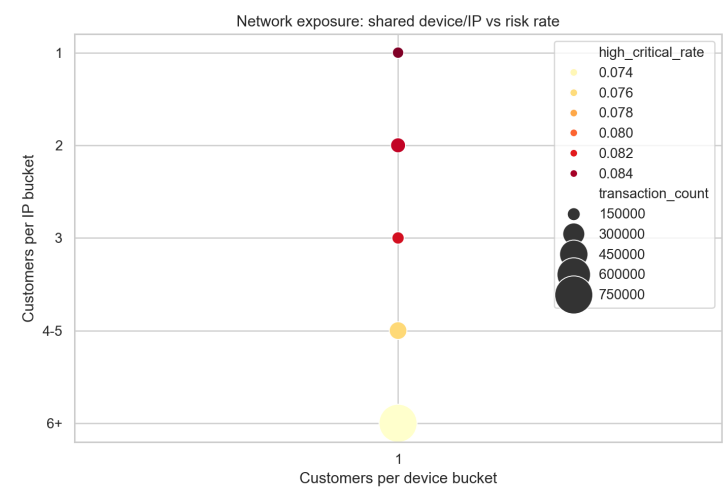
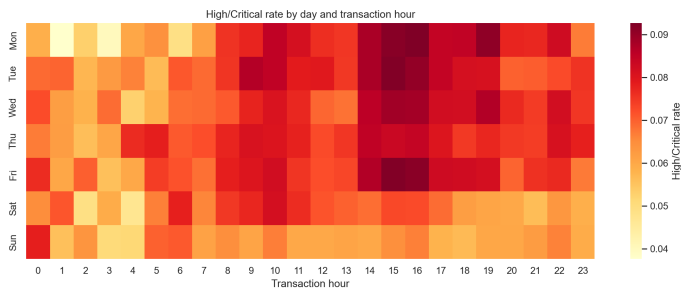
2019 monthly/quarterly backtest for review-rate stability



- Only 2019 data is available, so this is a temporal robustness check rather than a crisis-period validation.
- The KPI is not confirmed-fraud accuracy; it is weak-label prevention coverage, protected amount, and reason-code consistency over time.

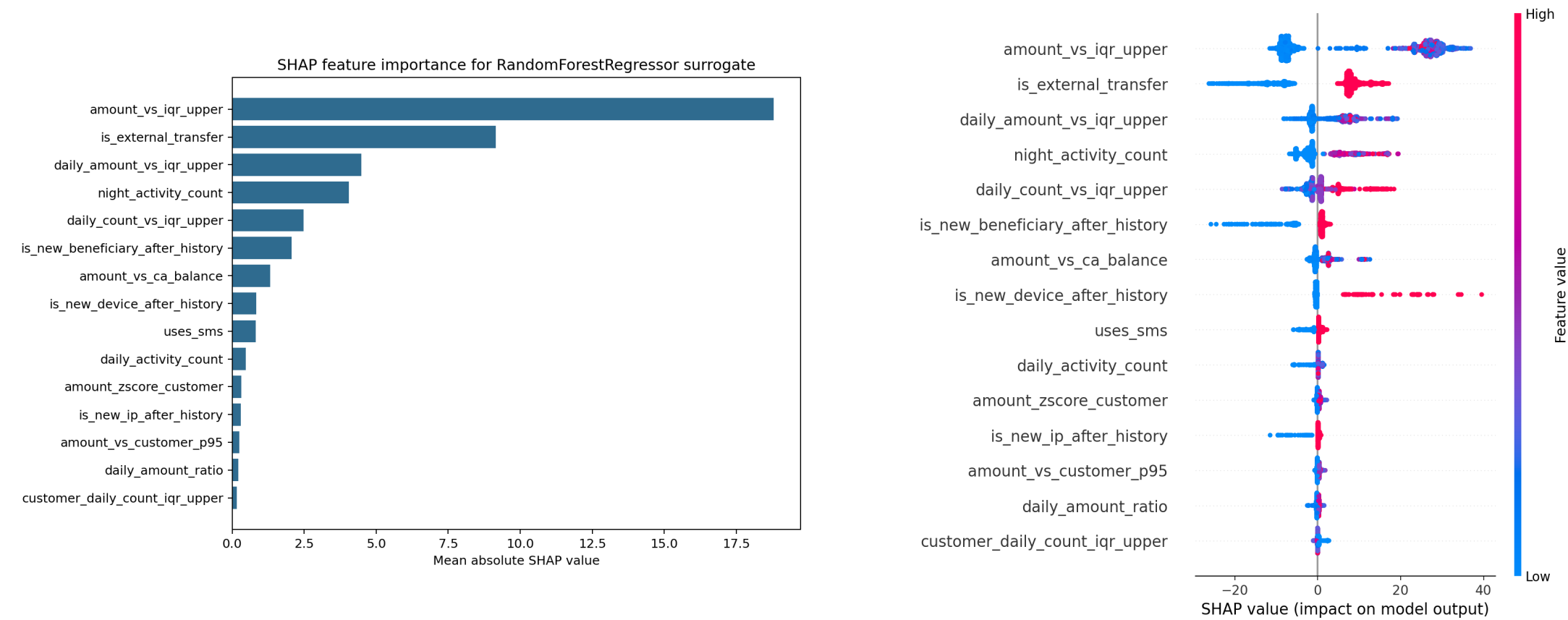
Risk Pattern Maps

Time, network and Customer 360 context explain where review capacity should go



xAI Engine

SHAP explains a tree surrogate of the supervised prevention score



Operational Recommendations

Turn model output into bank actions

- Critical: temporary hold, near-real-time manual review, customer verification, step-up authentication.
- High: step-up authentication before allowing the transaction to proceed.
- AML branch: network escalation rather than single-transaction review only.
- Medium: enhanced monitoring; escalate if repeated within the next review window.
- Feedback loop: investigator decisions become labels for Precision@K, Recall@K and supervised model tuning.

Live Demo

Streamlit advisor for transaction/customer review

- Run ``streamlit run src/demo_app.py``.
- Select a top-risk transaction or enter a CUSTOMER_NUMBER.
- The advisor returns risk band, root-cause branch, reason codes, recommended action and SHAP evidence.
- This is designed for a short judging demo or screen-recorded walkthrough.